MIL-C-24643/13B
3 October 1986
SUPERSEDING
MIL-C-0024643/13A(SH)
8 November 1985
MIL-C-24643/13
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 300 VOLTS, TYPE LSTTRS (INCLUDING VARIATION LSTTRSA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

This cable can be supplied in two variations: LSTTRS (unarmored) and LSTTRSA (armored).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, uncoated. AWG size 20, class B stranding.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene insulation. Nominal wall thickness 0.020 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. One black and one white or natural conductor for each pair.
Fourth - Two conductors, one black and one white or natural, cabled together to form a pair. The length of lay shall be such as to enable the completed cable to meet the twisting and bending endurance tests.
Fifth - Braided shield of AWG no. 34 or no. 36 copper, uncoated or tin coated. Braid angle 30 to 35 degrees; minimum coverage of 85 percent.
Sixth - Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, sealed. Standard identification code applied by method 2 on inner tape.

(B) denotes changes.

AMSC N/A FSC 6145
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Seventh - The required number of pairs (see table I) cabled together with left-hand lay. Length of lay shall be such as to enable the completed cable to meet the twisting and bending endurance tests. Cabling sequence shall be consecutive, starting with pair no. 1 from center outward. A single pair shall not be used as the central core of the cable. Fillers shall be used in the center and valleys, as necessary, to form a firm, round assembly.
Eighth - Binder tape applied helically with overlap.

Construction variant for type LSTTRS

Ninth - Cross-linked polyolefin jacket. Cable surface marking required.

Construction variant for type LSTTRSA

Ninth - Cross-linked polyolefin jacket.
Tenth - Braided metal armor and paint.

TABLE I. Details.

Military 1/ part no. M24643/13	Size 2/	Number of shielded pairs	Overall diameter (inches)			
			Type LSTTRS		Type LSTTRSA	
			min	max	min	max
-01UN	2	2	0.629	0.680	0.679	0.730
-02UN	4	4	.685	.740	.735	.790
-03UN	6	6	.814	.880	.864	.930
-04UN	8	8	.916	.990	.966	1.040
-05UN	10	10	.999	1.080	1.049	1.130
-06UN	12	12	1.017	1.100	1.067	1.150
-07UN	16	16	1.100	1.190	1.150	1.240

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration part number.
2/ Type LSTTRS and LSTTRSA.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet at 25°C, maximum...	11.95
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor.....	2000
Conductor to shield.....	1000
Shield to shield.....	200
Insulation resistance - megohms/1000 feet, minimum.....	500
Conductor continuity.....	No failure

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group A:	
Visual and dimensional	No failure
Mutual capacitance - pF/ft at 1 megahertz (MHz), maximum	28
Capacitance unbalance - percent, maximum	10
Characteristic impedance - ohms, at 1 MHz, minimum	75
maximum	90
Group B:	
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lbf/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lbf/in^2, minimum	1300
Elongation - percent, minimum	160
Twisting and bending endurance - (type LSTTRS only) - (see note 1)	
At 75 ± 2°C, cycles, minimum	6000
At minus 30 ± 2°C, cycles, minimum	2000

NOTE: 1. Roller diameters for bending endurance test shall be as follows:

Cable type and size	Roller diameter (inches, maximum)
LSTTRS-2	2.0
LSTTRS-4	2.2
LSTTRS-6	3.0
LSTTRS-8	3.5
LSTTRS-10	4.0
LSTTRS-12	4.0
LSTTRS-16	4.5

Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group C:	
Shrinkage	No failure
Permanence of printing (jacket) - (type LSTTRS only) cycles, minimum	250
Heat distortion - percent of unaged, maximum	30
Cable filler removability	No failure
Armor (LSTTRSA only) conformance to material, construction and coverage	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

	Aging and compatibility (cable)(125 ± 5°C)	No failure
(B)	Abrasion resistance (jacket) - scrapes, minimum	75
	Acid gas equivalent - percent, maximum	
	Jacket	2
	Fillers	2
	Insulation	18
	Halogen content - percent, maximum	
	Jacket	0.2
	Fillers	0.2
	Flame propagation (cable)	No failure
	Immersion (jacket)	
	Tensile strength - percent of unaged, minimum	50
	Elongation - percent of unaged, minimum	50
	Smoke index, maximum	
	Jacket	25
	Fillers	--
(B)	Insulation	45
	Toxicity index, maximum	
	Jacket	5
	Fillers	5
	Insulation	1.5
	Durometer (jacket) - (type A) hardness, minimum	80
	Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

All sizes 1500 feet (nominal)

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Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-1005-13)

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